

Use Case “Pay Order”

1. Use case code

UC002

2. Brief Description

This use case describes the interaction between AIMS software with the customer and VietQR when the customer wishes to pay the order

3. Actors

3.1 Customer

3.2 VietQr

4. Preconditions

Customer has entered valid delivery information

Aims software displayed the temporary invoice information for the order

5. Basic Flow of Events

1. Customer requests to pay order by QR code
2. AIMS software calls the get token API from VietQR
3. VietQR sends the access token to AIMS software
4. AIMS software calls the API of VietQR to generate the QR code (see Table 1)
5. VietQR sends the QR code for AIMS software
6. AIMS software displays the generated QR code along with the invoice information on the payment screen (see Table 3)
7. AIMS software calls the VietQR Test Callback API to simulate a successful payment transaction
8. VietQR calls AIMS’s Get Token API
9. AIMS returns an access token to VietQR
10. VietQR sends a payment callback to AIMS software through the Transaction Sync API (See Table 2)
11. AIMS software verifies the callback information
12. AIMS software confirms the payment result and updates the order status

6. Alternative flows

No	Location	Condition	Action	Resume location
1.	At Step 1	If the customer choose to pay order by credit card	AIMS Software inserts the use case “Pay order by credit card”	Use case ends

2.	At Step 6	If the callback information is invalid or cannot be verified	AIMS Software does not confirm the payment	Use case ends
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7. Input data

Table 1-Input data of VietQR code generation API (VietQR system)

No	Data fields	Description	Mandatory	Valid condition	Example
1.	BankCode	Code of the receiving bank account	Yes	Valid bank code supported by VietQR	MB
2.	BankAccount	Bank account number for the transfer	Yes	Valid bank account number	113000045678
3.	UserBankName	Name of the bank account holder	Yes	Must not contain Vietnamese accents	Nguyen Dang Quang
4.	Content	Payment transfer content	Yes	Maximum 19 characters, no special characters	PAYORDER01
5.	qrType	Type of VietQR code to generate	Yes	Dynamic QR = 0, Static QR = 1, Semi-dynamic QR = 3	0
6.	Amount	Total money of the corresponding media	No	- qrType = 0 or qrType = 3 required - Positive integer	175000

Table 2-Input data of Aims Transaction Sync API

No	Data fields	Description	Mandatory	Valid condition	Example
1.	BankAccount	Bank account number for the transfer	Yes	Valid bank account number	113000045678
2.	Amount	Total money of the corresponding media	Yes	- Positive integer	300000

3.	TransType	Payment transfer content	Yes	C = credit transaction, D = debit transaction	C
4.	Content	Payment transfer content	Yes		PAYORDER001
5.	Transaction ID	Total money of the corresponding media	No		TX123456
6.	Transaction Time	Timestamp when the payment occurred	No	Valid datetime format	16/03/2025 15:55:46
7.	OrderId	Identifier of the order	No	Maximum 19 characters, no special characters	ORDERID01

8. Output data

Table 3-Output data of QR code along with the invoice information (Aims software)

No	Data fields	Description	Display format	Example
1.	Bank Name	The name of the receiving bank		VietinBank
2.	Bank Account	Bank account number for the transfer		1130 0004 5678
3.	UserBankName	Name of the bank account holder		Nguyen Dang Quang
4.	Amount	Total money of the corresponding media	- Comma for thousands separator - Positive integer	150,000 VND
5.	QR Code	QR image generated from the string	QR Image	

9. Postconditions

- After a successful payment, the order is stored in the Aims Software with the status “*Pending Processing*”
- Aims software will display general information of the order and transaction information, and then send invoice with payment transaction information to the customer's email.

